**input · native-resolution OPG**



shared dispatcher - every image-consuming tool
runs on base_image; the crop chain never compounds

input geometry: TPU = canonical views + static
slots (no recompile); GPU = native so far -
decision pending; tools always run native

base_image · full resolution

## Policy

**Qwen3.5-9B**

reasoning node
thought + JSON tool calls

LoRA adapters:
SFT · GRPO (pluggable)

training

adapters

4

call

obs

## Tool node



live crop

**8 registered tools**

zoom · window · denoise
enhance · contrast
locate · nudge
bilateral compare · fdi

continuous parameters:
strength · center · width

on answer / budget

**final answer - findings JSON**

```
[{"quadrant":1,"tooth_position":6,"diagnosis":"Periapical Lesion","confidence":0.82}, ...]
```

5 per-image findings

## Evaluation

set-matched precision / recall / F1
over multi-finding predictions

95% bootstrap confidence intervals

heavy runs on GPU or TPU
(target decision pending - see input geometry)

**probes (mechanism evidence):**

- tool removal: with vs without
- perturbation: shown-box vs true-box
- nudge budget K sweep

(results pending, reported as open cells)

dev-split signal to training 6

training